Net filtration Pressure = Glomerular hydrostatic pressure − (Colloidal osmotic pressure + Capsular hydrostatic pressure).

Net filtration pressure = 55 mmHg − (30 mmHg + 15 mmHg)= 10mmHg

The effective glomerular pressure of 10 mmHg results in **ultrafiltration.** Glomerular filtration rate (GFR) is the volume of filtrate formed min^{-1} in all nephrons (glomerulus) of both the kidneys. In adults the GFR is approximately 120-125mL/min. Blood from the glomerulus is passed out through the efferent arteriole. The smooth muscle of the efferent arteriole contract resulting in vasoconstriction. Table 8.1 shows the relative concentrations of substances in the blood plasma and the glomerular filtrate. The glomerular filtrate is similar to blood plasma except that there are no plasma proteins. In cortical nephrons, blood from efferent arteriole flows into peritubular capillary beds and enters the venous system carrying with it recovered solutes and water from the interstitial fluid that surrounds the tubule.

Table 8.1 Concentration of substances in the blood plasma and in the glomerular filtrate

Substance	Concentration in blood Plasma/g dm^{-3}	Concentration in glomerular filtrate/g dm^{-3}
Water	900	900
Proteins	80.0	0.05
Aminoacids	0.5	0.5
Glucose	1.0	1.0
Urea	0.3	0.3
Uric acid	0.04	0.04
Creatinine	0.01	0.01
Inorganic ions (mainly Na^+, K^+ and Cl^-)	7.2	7.2

A person with cirrhosis of the liver has lower than normal levels of plasma proteins and higher than normal GFR. Explain why a decrease in plasma protein would increase GFR.

ii) Tubular Reabsorption

This involves movement of the filtrate back into the circulation. The volume of filtrate formed per day is around 170-180 L and the urine released is around 1.5 L per day, i.e., nearly 99% of the glomerular filtrate that has to be reabsorbed by the renal tubules as it contains certain substances needed by the body. This process is called selective reabsorption. Reabsorption takes place by the tubular epithelial cells in different segments of the nephron either by active transport or passive transport, diffusion and osmosis.

Proximal convoluted Tubule (PCT)- Glucose, lactate, amino acids, Na^+ and water in the filtrate is reabsorbed in the PCT. Sodium is reabsorbed by active transport through sodium- potassium (Na^+ K^+) pump in the PCT. Small amounts of urea and uric acid are also reabsorbed.

Descending limb of Henle's loop is permeable to water due the presence of aquaporins, but not permeable to salts. Water is lost in the descending limb, hence Na^+ and Cl^- gets concentrated in the filtrate.

Ascending limb of Henle's loop is impermeable to water but permeable to solutes such as Na^+, Cl^- and K^+.

The **distal convoluted tubule** recovers water and secretes potassium into the tubule. Na^+, Cl^- and water remains in the filtrate of the DCT. Most of the

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